

Tiago Pateiro

Senior Software Engineer — Backend & AI Systems

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SUMMARY

Backend-first engineer with 13+ years across product and consultancy. I work on PagerDuty's incubation team, building an agentic incident-response product with LangGraph, Python services, streaming APIs, and agent memory. Before that, I led Status Pages workstreams for a new team, from first commit to GA in six months and through enterprise upsell.

I write a design doc before code and ADRs for decisions worth revisiting. I lead workstreams across teams, mentor the engineers on them, and stay involved through launch and customer rollout.

Core stack: Python • .NET/C# • Elixir • Node.js • TypeScript • LangGraph • AWS • Terraform

EXPERIENCE

PagerDuty

Senior Software Engineer

CSOps Status team, then PD Advance / LABS (incubation).

Lisbon, Portugal

June 2022 – Present

AI incident response (2024 – present)

Backend and agent orchestration work on an agentic incident-response product built with LangGraph and Python services.

- Designed and led the metering service behind AI Actions, PagerDuty Advance's usage-based consumption model. It handles every metered call: over 2,000 requests per minute at roughly 50ms, keeping usage accounting accurate and per-account limits enforced.
- Built the shared memory layer where agents store context. Defined its contracts and use cases with the teams whose agents depend on it. It backs three production agents at roughly 500 requests per minute within a 120ms budget.
- Separated durable state from transient state in the LangGraph checkpointer. Resumed runs now replay persisted state without re-emitting intermediate tool traffic. This removed a recurring class of context-overflow failures.
- Worked on Agent Connectors, which link the SRE Agent to a customer's observability and knowledge sources over API or MCP so it can query their existing tooling.
- Shipped the Skills mechanism for the SRE Agent: custom instructions and runbooks it loads at run time to follow a team's procedures. Presented it internally at PagerDuty's AI showcase.

Status Pages (2022 – 2024)

- Project lead for subscriptions and notifications, two of the product's three critical user journeys and a flow central to pricing. Designed the notification service so channels could land incrementally, then shipped email, Slack, webhooks, and SMS across three consecutive quarterly releases, each on its committed date.
- Modelled the persistence schema, wrote the ADRs, and quantified an expected reduction in read latency from roughly 200ms to under 10ms up front.
- Built bring-your-own-identity-provider support for Private Status Pages, the premium enterprise tier. Customers can restrict access through OIDC providers including Azure, Okta, PingID, Salesforce, and JumpCloud with minimal setup. Cached JWKS and OpenID Connect discovery data in CloudFront to reduce calls to customer identity providers. The capability supported premium-tier upsell into enterprise accounts; I ran setup calls with those customers and wrote the provider configuration guides for the company knowledge base.
- Co-led custom-domain provisioning for DNS and email white-labelling, allowing notifications to send from the customer's domain. Diagnosed de-provisioning timeouts and moved the flow asynchronous through an ADR without moving the release date.
- Co-authored the product design document, led the Kubernetes migration, and bootstrapped the codebase for a new team. Ran the notification build-versus-buy analysis against the internal platform the team adopted.
- Owned observability, including SLIs and SLOs behind the team's SLAs, structured JSON logging with request correlation, and a monitoring overhaul that reduced on-call alert fatigue. Migrated the service to TypeScript and raised test coverage past 70%.

Stack: Python, Elixir, Node.js/TypeScript, LangGraph, AWS (Lambda, SQS, DynamoDB, CloudFront), Kubernetes, Terraform, MySQL, Redis, Elasticsearch

Defined.ai (formerly DefinedCrowd)

Senior Software Engineer → Lead

Lisbon, Portugal
July 2019 – May 2022

Speech and NLP training-data platform built on a global crowd workforce.

- Led a cross-functional team of eight: four backend engineers, two front-end engineers, two QA engineers, and a PM. Built automation and data tooling that replaced manual crowd workflows, reducing human handling per task and improving margin on delivered datasets.
- Owned the technical design for human-in-the-loop quality systems, contributor e-learning, and gamification. My scope included pre-screening tests for expert-level work. Contributors had to pass a language-specific assessment, such as an en-US test before translation tasks, before they could receive that work. Raised task throughput by 60%.

Stack: .NET, React, TypeScript, Azure, SQL Server, Spark, Kafka, Superset

Truphone

Software Engineer

Lisbon, Portugal
August 2017 – June 2019

Global eSIM provider and mobile network operator.

- Gathered stakeholder requirements and defined the architecture for customer-facing portals and mobile apps, on the product line responsible for over 60% of company revenue.
- Part of the team that shipped one of the first iPhone apps built on Apple's initial eSIM release, supporting over-the-air data plans and profile provisioning.

Stack: .NET, React Native, TypeScript, MongoDB, Elasticsearch, SQL, Java, Spark

BookedBy

Software Engineer

Lisbon, Portugal
January 2016 – August 2017

Booking and management SaaS for multi-location service businesses.

- Shipped an online booking product that ran standalone or embedded in client sites, removing the need for per-client custom development.
- Built a central management platform pushing promotions, inventory and users across multi-store chains, plus the data warehouse behind manager reporting.

Stack: Node.js, React, Backbone.js, .NET, PostgreSQL, Docker, AWS

EARLIER EXPERIENCE

ROFF — Developer (2012–2016). Designed an API integrating SAP QM with touchscreen shop-floor workflows and led technical direction for SAP integrations. Warehouse management with path-optimisation; automated production cost allocation for manufacturing clients. .NET, SQL, SAP ABAP.

InspirennovIT — Developer (2011–2012). PDF digital-signature workflow using the Portuguese citizen card, with SharePoint approval routing. .NET, SQL Server.

FCT-NOVA — Research Grant Holder (concurrent, 2011–2012). Academic data warehouse for candidate analytics; extraction tooling and geographic clustering. Java, Pentaho BI, MySQL.

TECHNICAL SKILLS

AI systems LangGraph, agent orchestration, tool calling, MCP, agent state and memory, prompt and context design, knowledge graphs

Programming Python, .NET/C#, Elixir, Node.js, TypeScript/JavaScript, SQL

Backend Distributed services, event-driven architecture, streaming APIs (SSE), OIDC/OAuth, design docs and ADRs

Cloud & infra AWS, Terraform, Kubernetes, Docker, CI/CD, observability (SLIs/SLOs, structured logging, dashboards)

Data MySQL, PostgreSQL, DynamoDB, Redis, Elasticsearch, Kafka, Spark

Languages Portuguese (native), English (fluent)

EDUCATION

MSc Knowledge Management & Information Systems — ISCTE-IUL, 2018

Postgraduate Diploma Computer Science — FCT-NOVA, 2012

BSc Computer Science — FCT-NOVA, 2010